Requirements Document

Landing Solutions

For Panther Cloud Air

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Introduction

Project Description & Project Goals

The intended outcome of Panther Cloud Air's proposed project is to build, test, and research the validity behind their airline operations. This will be done by developing a simulation that models the logistics and financial challenges of running an airline network. The simulation should encompass generating flights, booking tickets, tracking hours, airplane maintenance, fuel, and more. The simulation looks to test the routing airplanes must take, the hubs required to arrive and depart, but most importantly getting above airplane and airport costs. By taking in a holistic view of the airline operations in a simulation, this allows Panther Cloud Air to visualize performance and probability before having to commit to a large scale influx of capital.

This project also applies software engineering concepts to a real world problem. We are combining scheduling logic, simulation modeling and database integration in a team environment while building functional prototypes that can be used in the planning and decision making of airline operations.

Glossary

Term	Definition
UI	User Interface represents the visual component of any software, this is where the user makes a series of actions like scrolling, typing, clicking or swiping.
Database	The database is often used in an all encompassing way, but should be used to reference where all the user and system information is stored. Industry standard stores this information in a cloud, usually through a series of tables.
Server	The server refers to the series of system, functions and code that takes the input from the UI and sends it to the database to be stored and later used.
User Authentication	User Authentication is a broad term, but typically encompasses different processes or methodologies on validating if, when, and how a user can access a software.
Admin	Admin, short for Administrator, is a role to describe heightened permissioned. The permission tends to vary depending on the software, however they typically can manage accounts or software information or either.
Prototype	A prototype or minimum viable product is an early and often limited version of the software looking to be created. In modern times this is often done through mockups, not including any code.
Mockup / Wireframe	Mockup or Wireframes are a design tool when building software that allows for rapid revisions with a high degree of quality to present, discuss or visualize what a software will look like or do before building.
Version Control	A system like SourceTree used to track and manage changes made to code.

Appendices

Appendix A - UI Screenshots

Figure #1

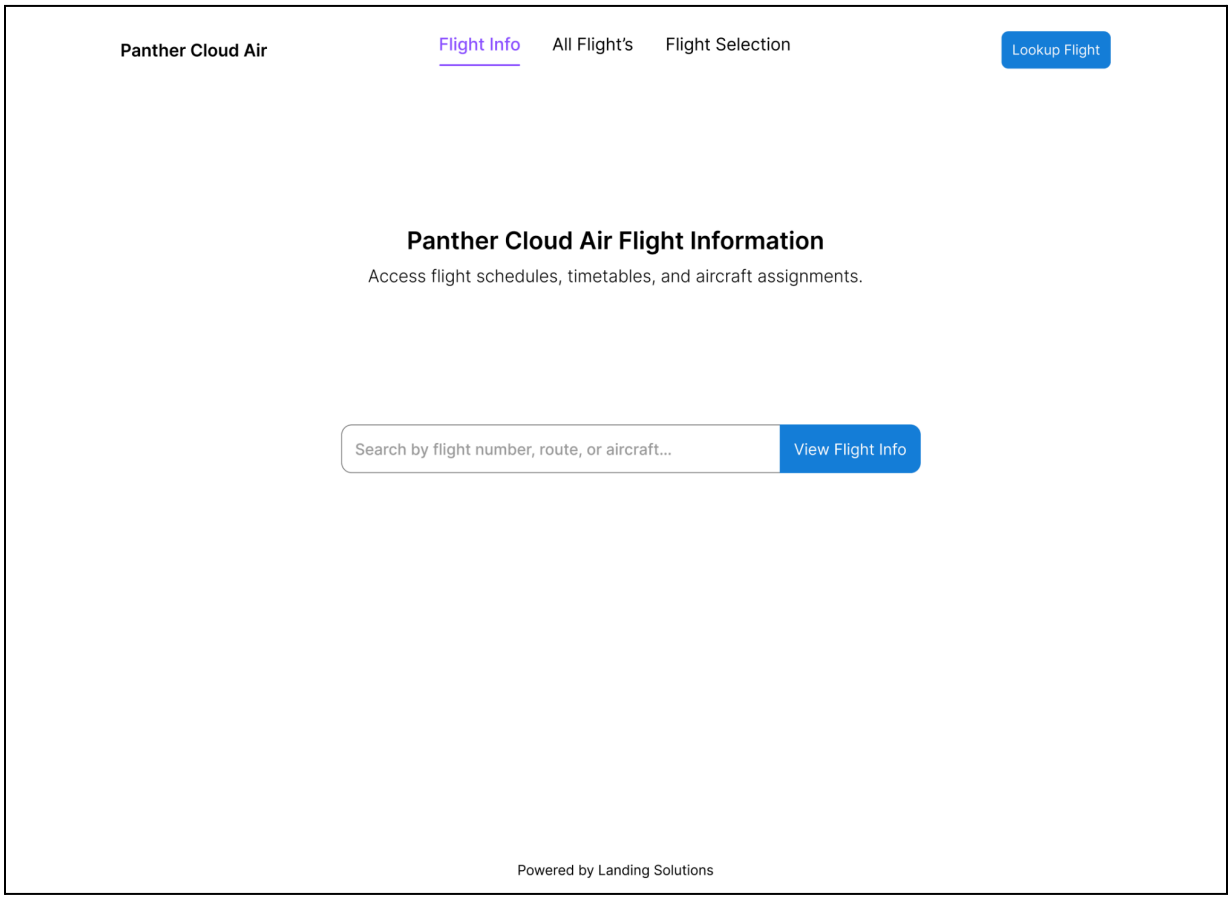
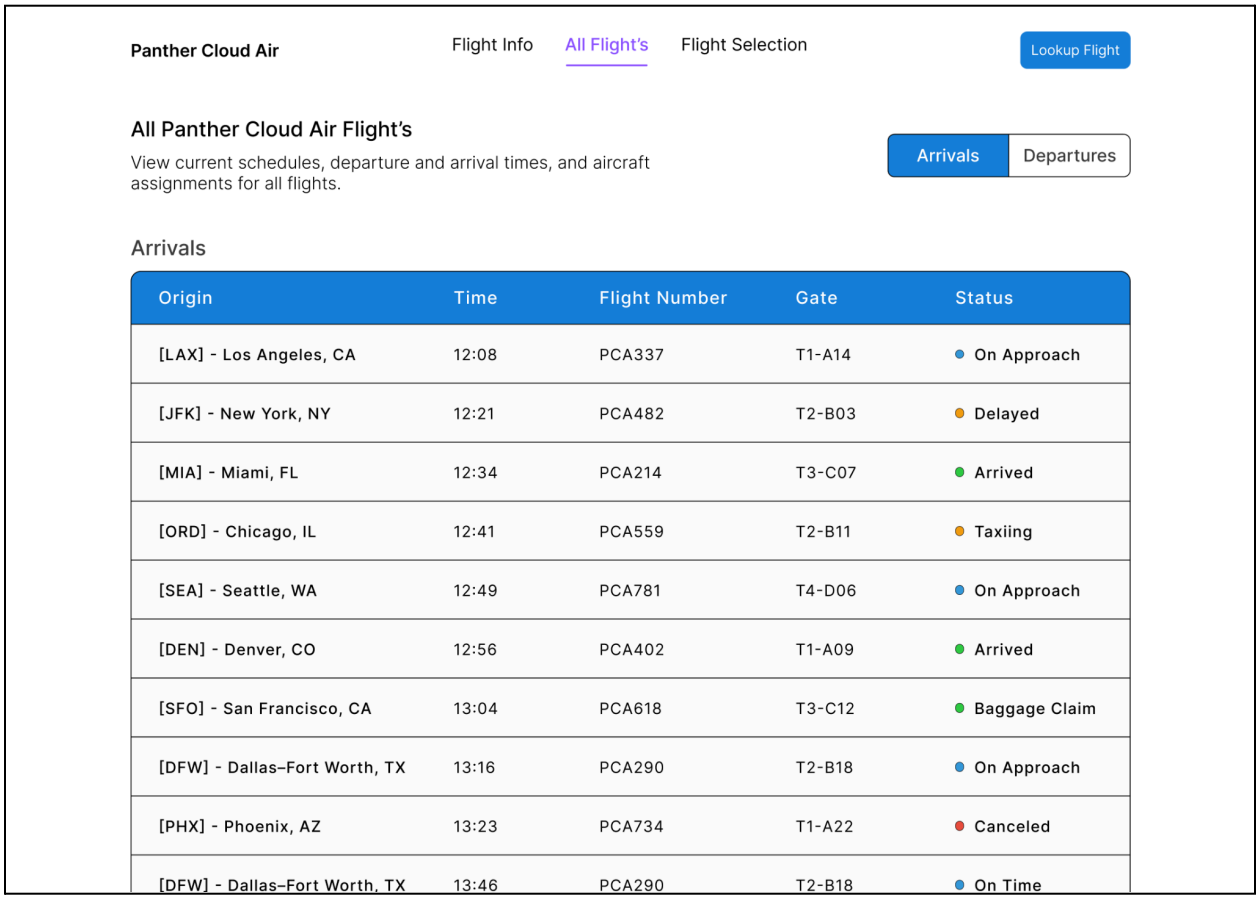


Figure 1 | Flight Info Screen | Airport Agents

This screen allows designated users to look up flights to provide information at airports.

Figure #2



Origin	Time	Flight Number	Gate	Status
[LAX] - Los Angeles, CA	12:08	PCA337	T1-A14	On Approach
[JFK] - New York, NY	12:21	PCA482	T2-B03	Delayed
[MIA] - Miami, FL	12:34	PCA214	T3-C07	Arrived
[ORD] - Chicago, IL	12:41	PCA559	T2-B11	Taxiing
[SEA] - Seattle, WA	12:49	PCA781	T4-D06	On Approach
[DEN] - Denver, CO	12:56	PCA402	T1-A09	Arrived
[SFO] - San Francisco, CA	13:04	PCA618	T3-C12	Baggage Claim
[DFW] - Dallas–Fort Worth, TX	13:16	PCA290	T2-B18	On Approach
[PHX] - Phoenix, AZ	13:23	PCA734	T1-A22	Canceled
[DFW] - Dallas–Fort Worth, TX	13:46	PCA290	T2-B18	On Time

Figure 2 | Flight Info Screen | Airport Agents

This screen allows designated users to easily see upcoming arrivals and departures.

Figure #3

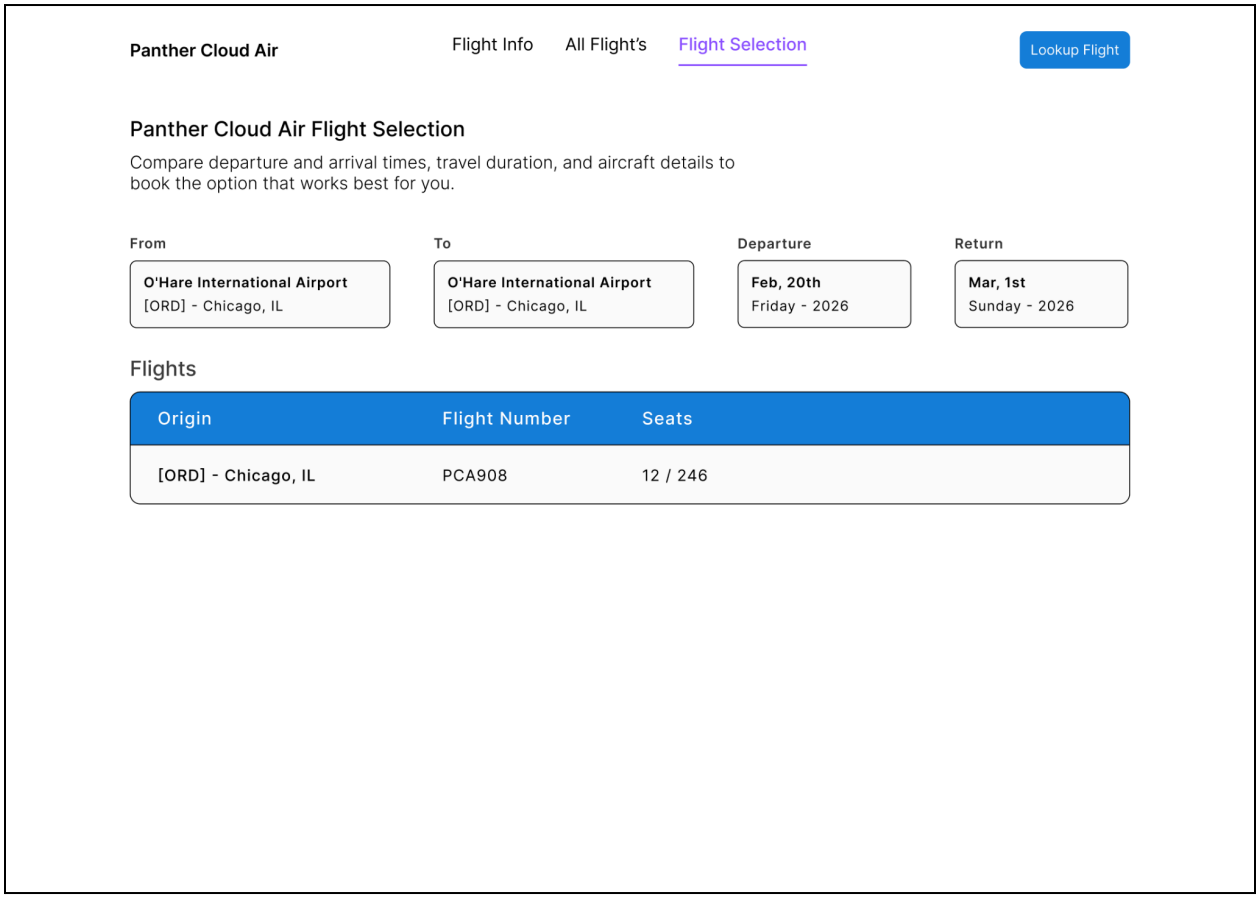


Figure 3 | Flight Info Screen | Airport Agents

This screen allows designated users to book a flight from and to an airport on given dates.

Users & Audience

Gate Agent

User Type:
Primary

Role:
This user provides useful information to passengers, book flights, and give flight status updates.

Characteristic:
- This user works at a given airport
- This user requires fast, reliable information regarding passengers and flights
- This user requires basic to intermediate computer skills

Pilot

User Type:
Primary

Role**:**
This user oversees flight operations, appropriate airplane maintenance and flight logging information.

Characteristic:
- This user operates airplanes at given airports
- This user requires fast and reliable information regarding airplane logs
- This user requires basic to intermediate computer skills

Admin

User Type:
Secondary

Role:
This user oversees simulated flight operations, appropriate airplane maintenance, as well as cost and operational efficiency.

Characteristic:

- This user operates from an officer or home
- This user requires fast and reliable information regarding active simulations
- This user requires intermediate to advanced computer skills

Functional Requirements

FR-01	Timetable Generation	The system automatically generates flight schedules for all 56 aircrafts. One of the key constraints is that routes have to be more than 150 miles apart, so we're not scheduling pointless short hops. It's not just about departure times. The system calculates accurate gate-to-gate flight times which means accounting for taxi time at each specific airport, takeoff, climb, descent, and landing. This precision matters because a small error can compound across dozens of flights and can create delays throughout the network.
FR-02	Hub Optimization	We need to identify 3 to 4 optimal hub airports based on passenger flow and geography. A big priority is minimizing overnight layovers in New York for passengers connecting to Paris. Nobody wants to sleep in JFK when they're trying to get to Paris. And the hub network has to support all-to-all connectivity, which is how major airlines operate. This is how we're building the same connectivity from day one as we can get from any airport to any other airport through the hubs, with at most one connection.
FR-03	Aircraft Fleet Management	Every aircraft needs scheduled maintenance every 200 flight hours, which takes about a day and a half at a hub. But here's the catch: we can only have 3 aircraft in maintenance at any given hub at the same time. Which is because of limited maintenance bays. So the system has to manage that scheduling conflict.
FR-04	Financial Operations	Tickets are priced at a 30% load factor, which means the flight is profitable even if only 30% of seats are filled. When we hit our goal of 75% load, we're making profit on every flight. The system handles USD to EUR conversion using live rates from xe.com as it is important for that daily Paris route where fuel costs are in euros. And it generates profit reports comparing revenue against all operating costs: fuel, airport fees, aircraft leases, everything.
FR-05	Fuel Calculations	The system figures out how much fuel each flight needs based on the distance and the type of aircraft being used. The Boeing 737-800 uses about 5.2 gallons of fuel per mile, while the Airbus A220-100 is a bit more efficient at around 4.2 gallons per mile showing that different planes burn fuel at different rates. It also keeps track of fuel prices, which are set at $6.19 per gallon in the United States and €1.97 per liter in Paris.
FR-06	14-Day Simulation with	The system runs a 14-day simulation of flight operations with different disruptions added along the way to see how well everything holds up. Day 1 is the baseline with no delays. On Day 3, weather hits and about 25% of

	Disruptions	flights get delayed anywhere from a minute to up to 15% of their flight time. Day 5 introduces icing, and on Day 7, a strong jet stream speeds up westbound flights by 12% but slows eastbound flights by the same amount. Day 9 adds gate delays affecting about 5% of flights, lasting between 5 and 90 minutes. Day 11 is the biggest disruption where a plane at a major hub completely fails. Finally, Day 13 simulates mass cancellations, where 8% of flights are canceled (passengers rebooked). The goal is to make sure the system can handle real-world disruptions.
FR-07	Flight Search	The system also includes a flight search tool for passengers. Users can search between any two airports, like JFK to LAX, and the system returns all possible itineraries. It shows departure/arrival times in the correct local time zones, layover airports and durations, total travel time, distance, and ticket price. International flights are clearly labeled, and time zone differences are handled automatically. Since this is what passengers use to book flights, the interface focuses on being easy to understand.
FR-08	Real-Time Aircraft Status	The system tracks where every aircraft in the fleet is along with the status. If you look up a plane by its tail number, you can see its location, how many hours it has flown, and when it's scheduled for its next maintenance check. Planes always start the day at the same airport where they finished the previous day, so the schedule stays realistic. This tracking helps dispatchers keep tabs on the fleet and also makes sure the same aircraft isn't accidentally scheduled for overlapping flights.
FR-09	User Authentication	The system uses secure login with access to protect data. Users must log in with a username and password. Failed attempts are logged, and after three failures, the account is temporarily blocked from attempting to log in again. This matters because the system contains sensitive business data.

Non-Functional Requirements

NFR-01	Performance Requirements	The system shall generate the full airline timetable and complete the required 14-day simulation in under 10 minutes on the target client's hardware will flight calculations accurate within 1 mile
NFR-02	Space Requirements	The database storing flights, passengers, airports and aircraft shall not exceed 2 GB of storage for a full two week simulation dataset
NFR-03	Usability Requirements	A user shall be able to access any primary system function (search flights, book a ticket, view financials) in no more than 3 menu selections from the main menu
NFR-04	Efficiency Requirements	The system shall return database query results (flight lookup, aircraft history, seat capacity) in under 1 second for at least 95% of queries
NFR-05	Dependability Requirements	The system shall continue executing the simulation if a single flight event fails and successfully complete at least 99% of scheduled events without terminating the run
NFR-06	Security Requirements	The system shall lock a user account for 5 minutes after three failed log-in attempts. This will iterate if failed logins continue, with time increasing to 15 min, then 30, 1 hour and 3 hours.
NFR-07	Environmental Requirements	The software shall run successfully on the client's machine without requiring platform-specific code changes
NFR-08	Operational Requirements	The system shall allow installation and execution with the provided setup instructions and complete full system setup in under 30 minutes on a clean machine
NFR-09	Development Requirements	The code shall compile successfully using the documented build process with zero build errors and no more than five warnings
NFR-10	Regulatory Requirements	The system shall follow FAA and industry standards including representing airports using the standards IATA airport codes and ensuring every aircraft has a unique tail number identifier
NFR-11	Ethical Requirements	The simulation shall accurately reflect the results of the simulation and shall not omit failed flights, cancellations or passenger disruptions to ensure a realistic product to the client and not restrict visibility of results
NFR-12	Legislative Requirements	All software tools and frameworks used in the system (such as React) shall be legally available for academic use and comply with respective licenses

NFR-13	Accounting Requirements	The simulation shall calculate all financial calculations for operation costs, revenue and profit/loss to two decimal places with rounding accuracy within $0.01

Customer

The customer is Panther Cloud Air, a new airline start-up company that seeks a system that simulates airline operations across a network of airports as a feasibility test/proof of concept for their business model. Panther Cloud Air requires the system to plan flight schedules, manage aircraft operations, evaluate real-world scenarios (such as delays, cancellations and passenger rebookings). This will allow the company to determine whether their operational strategy can run efficiently and at the end of the day, generate a profit for them and their stakeholders.

Operation Scenarios

Airport Attendee

In order for an airport attendee to view flight information, they must first log into the website. Once logged in, they will be able to view information regarding current flights, upcoming flights, and search for a specific flight. In order to book a new flight for a customer, they will search up that flight by filtering the upcoming flights with the customers preferred starting and final airport. Once the database returns flights matching those criteria, the flight attendant will be able to pick one and enter the customer's information. Upon successful completion of a booking, the booking information will become accessible to the customer who can further view and modify any flight information they have access to.

Customer

In order for a customer to view flight information, they must first log into the website. Once logged in, they will be able to view current flights for the day, view upcoming flights, view existing bookings, as well as book themselves onto a flight. In order to book a flight, a customer must put in required information such as preferred starting and final airport. Once the database returns flights matching those criteria and does not interfere with any existing flight plans, the customer may select any flight and complete the booking process.

Pilot

In order for a pilot to view flight information, they must first log into the website. Once logged in, they will be able to view current flights for the day, view upcoming flights, as well as view flight information such as seat counts and plane information. In order to see more detailed information, the pilot can go to their current flight and select what they want to view. The choices will include current plane statistics such as total flight hours, fuel amount, and if it will need maintenance soon. It will also show the flight statistics such as total flight time, departing and arriving airport, and gate information.

Goal hierarchy

Objective: Complete login window with security measures.

1. FR-09
    - Task: Establish a login portal with proper groupings (Employee, Pilot, Admin)
    - Task: Ensure after three bad passwords, user is unable to try again for a specified amount of time

Objective: Finalize employee window.

2. FR-01, FR-02, FR-03
    - Task: Allow employee users to access the timetable and search page
    - Task: Effectively display complete timetable from simulation seamlessly

Objective: Finalize pilot window.

3. FR-05, FR-06, FR-08
    - Task: Allow pilot users to access a plane oriented page
    - Task: Effectively search with plane information and display accurate flight simulation data

Objective: Finalize admin window.

4. FR-04, FR-05
    - Task: Allow admin users to access a profit management page
    - Task: Effectively display all cost related data in an organized manner

Objective: Successfully run errorless simulation and deliver complete documentation.

5. All FR & NFR
    - Task: Complete each deliverable and have it submitted to the github (User Manual, Specification Document, Final Design Document)

High-Level System Model

Figure #4

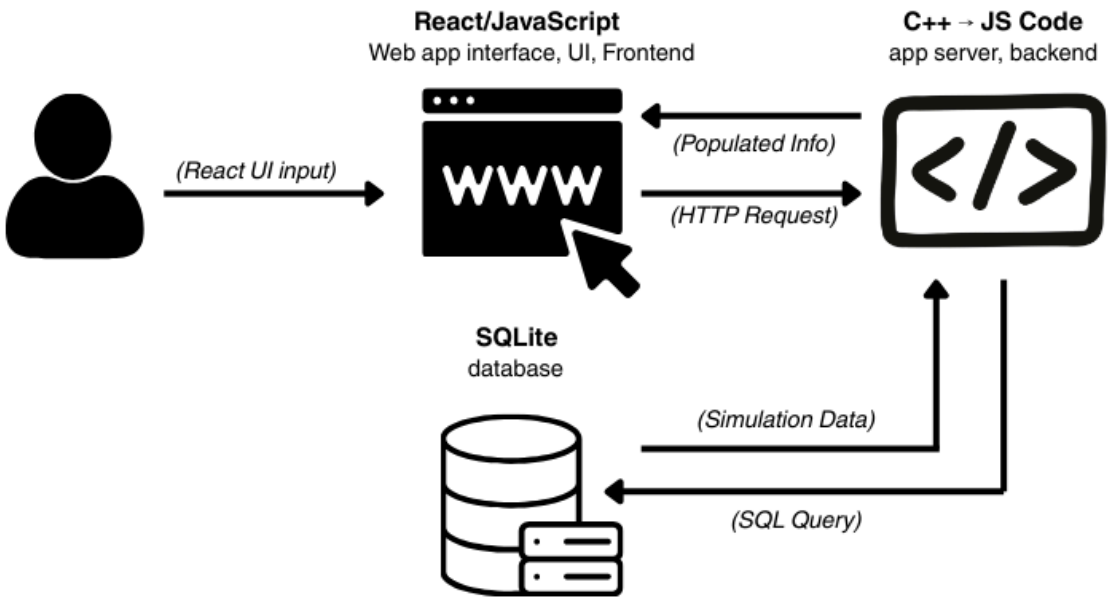


Figure 4 | High-Level System Model | Overall Application

This model displays a web application where our React/JavaScript frontend communicates with a JavaScript backend, which in turn reads and writes data in an SQLite database.

The user interacts with a React web interface that runs in the browser and collects input through the UI. This would be after initially logging in, and searching through our timetable, looking for flights, and other scenarios. When the user submits actions, React sends an HTTP request to our backend service.

The backend is a JavaScript server that used to be in C++, but we switched over to make connecting it to the database easier. This switch exposes HTTP endpoints. Once it receives the HTTP request, it runs the necessary application logic, and either queries or updates the SQLite database before sending populated information back to the React frontend as a response.

The SQLite database stores simulation or application data used by the backend.
The backend issues predetermined SQL queries to retrieve or persist this data, then packages the resulting simulation data into its HTTP responses so the React UI can display updated results to the user.

High-Level Cost Estimates

<u>Software Development Labor:</u>

- Team: 5 Developers
- Estimated Workload: 7 hours per week
- Project Duration: 12 weeks
- Estimated Payrate: $30/hour
- Total Hours: 420

Total Labor Cost: $12,600

<u>Equipment:</u>

- Model: Dell Pro 14 Premium PA 14250
    - Processor: Intel Core Ultra 7 268V, 2.2 GHz, 8 cores
    - Memory: 32 GB RAM
    - Storage: 120 GB disk space
    - Display: 14-inch Monitor
    - OS: Windows 11

Total Equipment Cost: $1659

Total Estimated Cost ~ $14,259